Shielded Payments for Decentralized Cloud Services: Anonymous Pay-Per-Use via Cross-Chain Shielded Pools, Prepaid Credits, and Rate-Limiting Nullifiers

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Abstract

We present a privacy-preserving payment system for decentralized cloud services—LLM inference, image generation, compute—where users pay anonymously and operators serve requests without learning the payer's identity. The system composes three layers: (1) a cross-chain UTXO-based shielded pool [1] with variable-amount JoinSplit transactions and multi-stablecoin wrapping, (2) a stateless gateway contract that bridges anonymous partial withdrawals to Tangle's Blueprint service lifecycle, and (3) a prepaid credit scheme where a single zero-knowledge proof funds an account from which many jobs are authorized via EIP-712 signatures. We describe the deployed system (Credit Mode) and a planned upgrade to per-request unlinkability via Rate-Limiting Nullifiers [3] composed with the existing audited VAnchor circuit (RLN Mode). The Credit Mode construction requires no modifications to Tangle's audited core contracts and introduces under 1,000 lines of new Solidity.

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1 Introduction

Every major AI inference provider—OpenAI, Anthropic, Google—requires account creation, identity verification, and traceable payment. A developer querying an LLM for code review, legal analysis, or medical information creates a permanent record linking their identity to every prompt. Decentralized alternatives inherit the same problem: on-chain payments link wallet addresses to service consumption with full public auditability.

This is not hypothetical. Inference providers can profile users by query content, timing, and volume. Employers can monitor which employees use AI assistance. Competitors can observe which organizations are training custom models. The payment trail is the surveillance vector.

Buterin and Crapis [2] proposed *ZK API Usage Credits*: deposit into a smart contract, generate a zero-knowledge proof per API call to authorize payment without revealing identity. The proposal achieves strong privacy—per-request unlinkability via Rate-Limiting Nullifiers—but requires a Groth16 proof per API call (∼640ms generation, ∼300k gas on-chain).

We present a two-phase approach that ships immediately (Credit Mode) while providing a clear upgrade path to per-request unlinkability (RLN Mode):

Credit Mode (deployed): One ZK proof funds a pseudonymous credit account. Many ECDSA signatures authorize individual job payments (∼50k gas each). Requests within a credit session share a pseudonym; sessions are unlinkable.

RLN Mode (planned): Each request carries its own RLN-based ZK proof, verified off-chain by the operator. Zero on-chain transactions per request. Full per-request unlinkability. The RLN circuit [3] composes with the audited VAnchor circuit [1, 5] with ∼100 new constraints.

1.1 Contributions

1. A UTXO-based privacy model with partial withdrawals producing unlinkable change notes, enabling incremental spending from a single deposit.
2. A two-phase payment scheme: one ZK proof amortized over many ECDSA-signed payments (Credit Mode), upgradeable to per-request RLN proofs verified off-chain (RLN Mode).
3. A gateway architecture bridging VAnchor shielded pools to Tangle's service lifecycle with zero core protocol modifications.
4. Multi-asset wrapping (USDC, USDT, DAI → single pool token) maximizing the anonymity set.
5. A cross-chain bridge via LayerZero V2 with dynamic edge addition, supporting deposits on 8 chains.
6. An audit-preserving circuit composition strategy where the RLN Mode upgrade reuses audited sub-circuits (VAnchor [5], RLN [4]) with ∼100 lines of new glue constraints.

2 Preliminaries

We adopt notation from the Webb Protocol [1]. Let $H : \mathbb{F}_p^* \to \mathbb{F}_p$ denote Poseidon [9], a zkSNARK-friendly hash over the BN254 scalar field.

Definition 1 (UTXO Commitment)**.** A commitment to a UTXO with chain identifier i, value v, public key $\mathsf{pk} = H(\mathsf{sk})$, and blinding factor $r \xleftarrow{\$} \mathbb{F}_p$ is:

$$\mathsf{cm} = H(i, v, \mathsf{pk}, r)$$

Definition 2 (Nullifier)**.** The nullifier for commitment cm at Merkle path index j, with hash-based signature $\sigma = H(\mathsf{sk}, j, \mathsf{cm})$, is:

$$\mathsf{sn} = H(\mathsf{cm}, j, \sigma)$$

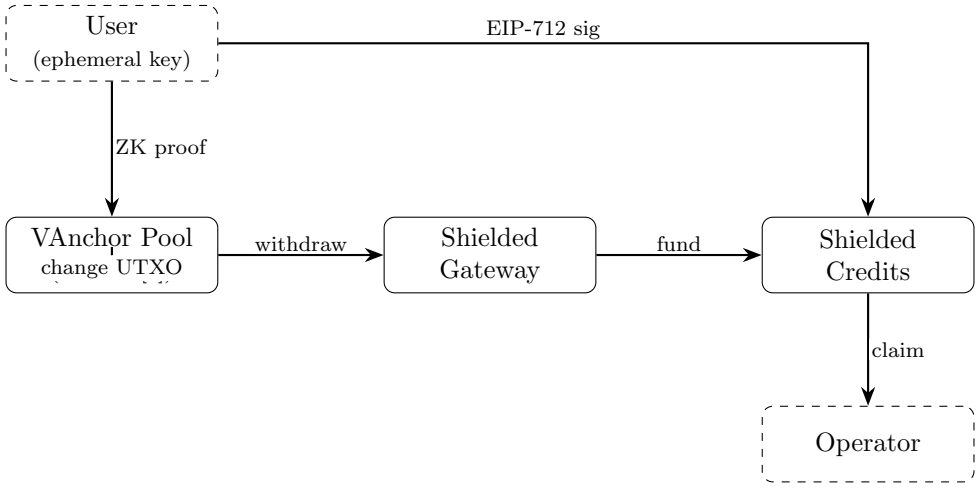
Binding σ to the secret key prevents nullifier grinding by parties who observe commitments but do not hold sk.

Definition 3 (Cross-Chain Merkle Membership)**.** Given Merkle roots $\mathbf{MR} = \{\mathsf{MR}_1, \ldots, \mathsf{MR}_k\}$ from k connected anchor instances, commitment cm satisfies membership if:

$$\mathsf{reconstruct}(\mathsf{cm}, \mathsf{path}(\mathsf{cm}), H) \in \mathbf{MR}$$

The circuit is compiled with a fixed k (currently $k = 8$). Unoccupied edge slots are filled with a deterministic zero hash. New chains are added on-chain via `addEdge` without circuit recompilation or redeployment.

3 System Architecture (Credit Mode)



3.1 UTXO Model and Partial Withdrawals

The VAnchor [1] implements a UTXO-based shielded pool with JoinSplit transactions (2-in-2-out or 16-in-2-out). Unlike fixed-denomination mixers (e.g., Tornado Cash [6]), each UTXO carries an arbitrary value, enabling partial withdrawals:

1. User holds $\mathsf{UTXO}(v)$ in the pool (e.g., $v = 100$ USDC).
2. User generates a Groth16 proof spending $\mathsf{UTXO}(100)$ with outputs:
   - Withdrawal of 10 to the gateway ($\mathsf{extAmount} = -10$)
   - Change $\mathsf{UTXO}(90)$ re-shielded in the pool
3. The circuit enforces conservation: $\mathsf{public_amount} + \sum v_{\mathsf{in}} = \sum v_{\mathsf{out}}$.
4. The spent UTXO's nullifier is marked on-chain. The change UTXO has a fresh commitment with fresh blinding—**unlinkable to the original deposit or prior withdrawals**.

The user repeats this process for each credit session. Each withdrawal consumes a different UTXO, produces a different nullifier, and creates a different change commitment. An observer sees n withdrawals from the pool but cannot determine whether they originate from one depositor or n different depositors.

3.2 Shielded Gateway

The `ShieldedGateway` is a stateless proxy between the VAnchor and Tangle's service layer. It executes a VAnchor withdrawal (validating the Groth16 proof and marking nullifiers on-chain), receives tokens, and atomically forwards them to one of:

- `shieldedRequestService`: Create a Tangle service instance with the gateway as owner and ephemeral addresses as permitted callers.
- `shieldedFundService`: Top up an existing service's escrow.
- `shieldedFundCredits`: Fund a prepaid credit account (Section 3.3).

Property 1 (Atomicity)**.** The gateway holds no funds beyond a single transaction. Token receipt and forwarding occur in the same call. Failure at any point reverts the entire transaction, including the VAnchor withdrawal.

Property 2 (Non-Linkability)**.** The gateway stores no persistent state mapping nullifiers to service or credit identifiers. The on-chain trace reveals only the flow `VAnchor` → `Gateway` → `Credits`, not which depositor initiated it.

3.3 Shielded Credits

The core Credit Mode contribution: a prepaid credit system that amortizes one ZK proof across many service invocations.

Definition 4 (Credit Account)**.** Identified by $\mathsf{C} = \mathsf{keccak256}(\mathsf{pk}_s \| \mathsf{salt})$ where pk_s is an ephemeral ECDSA public key and $\mathsf{salt} \xleftarrow{\$} \{0,1\}^{256}$. Account state: $(\mathsf{pk}_s, \mathsf{token}, \mathsf{balance}, \mathsf{nonce})$.

Definition 5 (Spend Authorization)**.** An EIP-712 [8] typed signature over:

$$\mathsf{SpendAuth} = \mathsf{Sign}_{\mathsf{sk}_s}\big(\mathsf{C},\ \mathsf{serviceId},\ \mathsf{jobIndex},\ \mathsf{amount},\ \mathsf{operator},\ \mathsf{nonce},\ \mathsf{expiry}\big)$$

The designated $\mathsf{operator}$ is the only address that can claim payment. Claims after expiry are rejected; unclaimed funds can be reclaimed by anyone via ReclaimExpiredAuth.

The credit lifecycle proceeds as: FundCredits (pull tokens, initialize account) → AuthorizeSpend (verify signature, deduct balance, store pending claim) → ClaimPayment (transfer to operator, require $\mathsf{msg.sender} = \mathsf{operator}$ and $\mathsf{timestamp} \leq \mathsf{expiry}$).

4 Privacy Analysis

4.1 Operator Knowledge Boundary

Information	Operator	Chain observer
User's wallet address	×	×
Deposit amount / total shielded balance	×	×
Which deposit funded this credit account	×	×
Credit commitment C	✓	✓
Payment amount per job	✓	✓
Request content (prompt, image, query)	✓	×
Linkage between jobs within same C	✓	✓
Linkage between different credit accounts	×	×

The operator knows *what* was requested and *how much* was paid, but not *who* is paying. This is the "cash at a store" model: the merchant sees the transaction but cannot trace it to a bank account.

4.2 Unlinkability Across Sessions

When credit account C_1 is depleted, the user generates a new partial withdrawal from their shielded UTXO, creating C_2 with a fresh ephemeral key. The two accounts are cryptographically unlinkable because they use different ephemeral keys, different commitments, different nullifiers, and the connecting change UTXO has fresh blinding. The anonymity set is the total number of depositors in the pool.

4.3 Limitations of Credit Mode

Intra-session linkability. Multiple spends from the same C are linkable—the commitment is a pseudonym. Full per-request unlinkability requires a ZK proof per request (addressed in RLN Mode, Section 5).

Operator sees request content. Payment privacy does not imply computation privacy. TEE-based execution would address this orthogonally.

Anonymity set dependence. Privacy degrades with few depositors. Multi-asset wrapping and cross-chain deposits are designed to grow the set.

No IP-level privacy. Network-layer anonymity (Tor, Nym) is orthogonal and composable.

5 Per-Request Unlinkability via Rate-Limiting Nullifiers (RLN Mode)

The Credit Mode system links requests within a credit session—the commitment C is a pseudonym shared across all requests from the same account. For sensitive workloads (medical queries, legal analysis, investigative journalism), even pseudonymous linkability is unacceptable. The RLN Mode protocol eliminates intra-session linkability entirely: every request is cryptographically independent, with zero on-chain transactions per request.

5.1 Definitions

Definition 6 (Epoch)**.** An epoch e is a monotonically increasing index that partitions time into rate-limiting windows. A user may issue at most one request per epoch. Epochs may correspond to block ranges, time intervals, or application-defined windows.

Definition 7 (RLN Identity)**.** A user's RLN identity is derived from their VAnchor UTXO commitment:

$$\mathsf{identitySecret} = \mathsf{sk}_{\mathsf{vanchor}}$$

$$\mathsf{identityCommitment} = H(\mathsf{identitySecret})$$

This binds the RLN identity to the shielded pool deposit without introducing a new secret.

Definition 8 (External Nullifier)**.** For epoch e and application identifier rlnId:

$$\mathsf{extNull} = H(e, \mathsf{rlnId})$$

The rlnId prevents cross-application nullifier collisions.

Definition 9 (Rate-Limiting Nullifier and Share)**.** For a request with message m in epoch e:

$$\mathsf{rlnNull} = H(\mathsf{identitySecret}, \mathsf{extNull}) \tag{1}$$

$$a = H(\mathsf{identitySecret}, \mathsf{extNull}) \tag{2}$$

$$x = H(m) \tag{3}$$

$$y = \mathsf{identitySecret} + a \cdot x \pmod{p} \tag{4}$$

The tuple $(\mathsf{rlnNull}, x, y)$ is revealed with the proof. A single share (x, y) is information-theoretically insufficient to recover $\mathsf{identitySecret}$. Two distinct shares (x_1, y_1) and (x_2, y_2) from the same epoch yield:

$$\mathsf{identitySecret} = \frac{y_2 - y_1}{x_2 - x_1} \pmod p$$

Definition 10 (Refund Ticket)**.** After serving a request, the operator signs a refund ticket:

$$r = \big(C_{\max} - C_{\mathsf{actual}},\ \sigma_{\mathsf{server}}\big)$$

where $C_{\max}$ is the maximum per-request cost committed in the proof, C_{actual} is the realized cost, and σ_{server} is the operator's ECDSA signature over the refund value. The user accumulates refund tickets locally. Their total unlocks additional request capacity via the solvency constraint.

5.2 Composed Circuit

The composed circuit reuses two audited sub-circuits verbatim, adding only glue logic.

Construction 1 (VAnchor-RLN Payment Circuit)**.** For public inputs:

$$x = \{\mathsf{public_amount}, \mathbf{MR}, i, \{\mathsf{sn}_k\}, \{\mathsf{cm}_o\}, \mathsf{rlnNull}, x_{\mathsf{share}}, y_{\mathsf{share}}, e, \mathsf{extDataHash}\}$$

and private witness:

$$\mathbf{w} = \mathbf{w}_{\mathsf{vanchor}} \cup \{\mathsf{identitySecret}, \mathsf{epochIndex}, C_{\max}, R, \{\mathsf{refundSigs}\}\}$$

the relation $\mathcal{R}$ enforces:

Sub-circuit A (VAnchor, ∼38k constraints, audited [5]):

$$\text{UTXO commitments, nullifiers, Merkle membership, amount conservation} \tag{5}$$

Outputs the input UTXO values $\{v_k\}$ and the user's secret key sk.

Glue (∼80 constraints, **new**):

$$\mathsf{identitySecret} = \mathsf{sk} \tag{6}$$

$$(\mathsf{epochIndex} + 1) \times C_{\max} \leq \sum v_k + R \tag{7}$$

$$\mathsf{extNull} = H(e, \mathsf{rlnId}) \tag{8}$$

Equation (6) binds the RLN identity to the VAnchor secret key—no new key material. Equation (7) ensures the user has deposited enough (plus accumulated refunds R) to cover their next request at cost $C_{\max}$. Equation (8) computes the external nullifier from the epoch.

Sub-circuit B (RLN, ∼5k constraints, audited [4]):

$$\mathsf{rlnNull} = H(\mathsf{identitySecret}, \mathsf{extNull}) \tag{9}$$

$$a = H(\mathsf{identitySecret}, \mathsf{extNull}) \tag{10}$$

$$y_{\mathsf{share}} = \mathsf{identitySecret} + a \cdot x_{\mathsf{share}} \pmod p \tag{11}$$

Total new constraints: ∼80 (3 Poseidon hashes + 1 comparison + 1 multiplication). The sub-circuit constraints are unchanged—existing audit findings apply without modification.

5.3 Protocol

Algorithm 1 RLN Mode: Per-Request Unlinkable Payment

Setup (one-time, on-chain):
User deposits stablecoins into VAnchor pool, obtaining $\mathsf{UTXO}(v)$
User registers $\mathsf{identityCommitment} = H(\mathsf{sk})$ in the RLN settlement contract

Per-request (off-chain, ∼640ms):
$e \leftarrow$ current epoch
$m \leftarrow H(\mathsf{request_payload})$
$\pi \leftarrow \textsc{Prove}(\mathsf{pp}, x, \mathbf{w})$ ▷ Composed VAnchor-RLN proof
User sends $(\pi, \mathsf{rlnNull}, x_{\mathsf{share}}, y_{\mathsf{share}}, e, \mathsf{request})$ to operator

Operator verification (off-chain, ∼5ms):
$\textsc{Verify}(\mathsf{pp}, x, \pi) \stackrel{?}{=} 1$
Check $\mathsf{MR}_{\mathsf{proof}} \in$ recent on-chain Merkle roots
Check $\mathsf{rlnNull} \notin \mathsf{UsedNullifiers}$
$\mathsf{UsedNullifiers} \leftarrow \mathsf{UsedNullifiers} \cup \{\mathsf{rlnNull}\}$
Store $(\mathsf{rlnNull}, x_{\mathsf{share}}, y_{\mathsf{share}})$ for slashing detection
Serve request, compute C_{actual}
Sign refund ticket: $r \leftarrow (C_{\max} - C_{\mathsf{actual}}, \sigma_{\mathsf{server}})$
Return $(\mathsf{response}, r)$ to user

Batch settlement (on-chain, periodic):
Operator calls $\textsc{BatchClaim}(\{\mathsf{rlnNull}_i, C_{\max,i}\}_{i=1}^{n}, \mathsf{operator})$
Contract marks nullifiers as used, transfers $\sum C_{\max,i}$ to operator

Slashing (on-chain, permissionless):
if two shares $(x_1, y_1), (x_2, y_2)$ exist for the same $\mathsf{rlnNull}$ **then**
    $\mathsf{sk} \leftarrow (y_2 - y_1) \cdot (x_2 - x_1)^{-1} \pmod p$
    Verify $H(\mathsf{sk}) = \mathsf{identityCommitment}$
    Transfer user's deposit to slasher
end if

5.4 Refund Accumulation

The solvency constraint (Eq. 7) gates the number of requests a user can make. Without refunds, a user with deposit D and per-request cap $C_{\max}$ can make at most $\lfloor D/C_{\max} \rfloor$ requests. With refund accumulation, unused capacity is returned:

After request i, the user holds refund tickets $\{r_1, \ldots, r_i\}$ with total value $R = \sum_{j=1}^{i}(C_{\max} - C_{\mathsf{actual},j})$. The solvency constraint for request $i+1$ requires:

$$(i+1) \cdot C_{\max} \leq D + R$$

Since $R = i \cdot C_{\max} - \sum C_{\mathsf{actual},j}$, this simplifies to:

$$C_{\max} \leq D - \sum_{j=1}^{i} C_{\mathsf{actual},j}$$

The user can continue making requests as long as their cumulative *actual* cost does not exceed their deposit—regardless of how many requests they make or how large $C_{\max}$ is.

The circuit verifies the refund total R by checking the operator's signatures over individual refund values inside the ZK proof. This adds ∼20 constraints per accumulated refund ticket (one Poseidon hash + one ECDSA verification per ticket, batched via a Merkle root of refund tickets).

5.5 Proof Pre-Generation and Parallelism

The 640ms proof generation time is CPU-bound and embarrassingly parallelizable. The SDK maintains a pool of pre-generated proofs; the user never blocks on proof generation during a request. On an 8-core machine, 8 proofs complete in ∼640ms wall time, yielding an effective throughput of ∼12.5 proofs/second.

The pre-generation strategy is straightforward: the SDK observes the user's remaining epoch budget and speculatively generates proofs for future epochs in background threads. When a request arrives, the SDK pops a pre-generated proof from the pool and immediately sends it to the operator. If the pool is empty (cold start), the user experiences one ∼640ms delay before the pipeline reaches steady state.

This transforms the per-request latency profile from ∼640ms (sequential proof generation) to near-zero (proof pool hit), at the cost of speculative CPU usage that would otherwise be idle. The operator-side verification remains ∼5ms regardless.

5.6 Settlement Contract

The `RLNSettlement` contract (∼200 lines) manages deposits, nullifiers, and slashing. It does *not* verify ZK proofs on-chain—that is the operator's responsibility (backed by staking/slashing in Tangle's core protocol).

- `deposit(token, amount, identityCommitment)`: User locks funds against their identity commitment.
- `batchClaim(nullifiers[], amounts[], operator)`: Operator submits accumulated request nullifiers and claims payment in bulk. Each nullifier is marked as used; duplicate nullifiers revert.
- `slash(nullifier, x1, y1, x2, y2, identityCommitment)`: Anyone submits two distinct Shamir shares with the same nullifier. The contract recovers sk, verifies $H(\mathsf{sk}) = \mathsf{identityCommitment}$, and transfers the user's deposit to the slasher.
- `withdraw(identityCommitment, amount)`: User reclaims remaining deposit (minus claimed amounts).

5.7 Privacy Comparison

Property	**Credit Mode**	**RLN Mode**	**Notes**
Depositor unlinkability	✓	✓	VAnchor provides both
Inter-session unlinkability	✓	✓	Fresh keys / fresh nullifiers
Intra-session unlinkability	×	✓	**Key difference**
Per-request on-chain tx	∼50k gas	0 (batched)	RLN Mode settles periodically
Per-request client latency	<100ms	∼640ms	RLN Mode generates ZK proof
Refund for unused capacity	×	✓	Server-signed tickets
Abuse protection	Nonce ordering	Shamir slashing	RLN Mode is cryptoeconomic
New audit surface	1,000 LOC Sol.	∼80 constraints	RLN Mode reuses audited circuits

Both modes coexist on the same operator. Users choose their privacy level per request: Credit Mode for low-latency pseudonymous access, RLN Mode for full per-request unlinkability.

6 Use Cases

The system targets workloads where *usage patterns* are as sensitive as the data itself.

LLM Inference. Developers querying LLMs for code review, legal analysis, or medical triage create prompt histories that reveal strategic intent. Shielded payments decouple the query from the querier.

Image and Video Generation. Creative professionals generating images or video may wish to keep their prompts and creative direction private from the service provider and from on-chain observers.

AI Agent Execution. Autonomous agents paying for inference, tool use, and compute should not reveal their controller's identity. The agent's on-chain address is decoupled from the entity that funds it.

Fine-Tuning. Organizations submitting proprietary datasets for model fine-tuning can prevent competitors from monitoring training activity via payment trails.

General Cloud Compute. Any Blueprint-based service—databases, CI/CD, storage, web hosting—can accept shielded payments. The credit system is service-agnostic.

7 Cross-Chain Architecture

The VAnchor supports cross-chain deposits via linked Merkle roots relayed through LayerZero V2. A user deposits USDC on Ethereum; the root is relayed to Base; the user withdraws on Base citing the Ethereum root in the membership proof.

The circuit compiles with $\mathsf{maxEdges} = 7$ (8 chains including self). Unoccupied edge slots are filled with deterministic zero hashes. Chains are added dynamically via `addEdge`—a single multisig transaction that sets the chain mapping, LayerZero peer, and initial Merkle root atomically.

Initial deployment: Ethereum, Arbitrum, Base, Hyperliquid, BSC. Room for 3 additional chains without circuit recompilation.

8 Cost Analysis

Approach	Per-request gas	Per-request latency	Privacy
ZK proof per call [2]	∼300k (on-chain)	∼640ms	Full
VAnchor withdrawal per call	∼300k (on-chain)	∼640ms	Full
Credit Mode (Prepaid Credits)	**∼50k (on-chain)**	**<100ms**	**Pseudonymous**
RLN Mode (Per-Request Proofs)	**0 (batched)**	**∼640ms**	**Full**
x402 [7] (no privacy)	∼30k (on-chain)	<50ms	None

The Credit Mode one-time proof cost is amortized over all jobs in a session. A \$50 credit account at \$0.10/call amortizes over 500 calls. The RLN Mode proof cost is per-request but entirely client-side with zero on-chain gas.

9 Implementation

The Credit Mode system is deployed as a standalone Foundry project at `tangle-network/shielded-payment-g` It compiles against the audited protocol-solidity contracts (git submodule, unmodified) and requires no changes to Tangle's core protocol.

Component	Lines	Audit status
VAnchor + TokenWrapper + Merkle	∼3,000	Veridise [5]
RLN circuit (RLN Mode)	∼500	yAcademy [4]
OpenZeppelin 4.x / 5.x	—	OpenZeppelin
ShieldedGateway	188	New (Credit Mode)
ShieldedCredits	232	New (Credit Mode)
LayerZeroAnchorBridge	260	New (Credit Mode)
Interfaces	252	New (Credit Mode)
Glue constraints (RLN Mode)	∼100	Planned

Test coverage: 65 Forge tests (Credit Mode, including 4 fuzz tests for financial invariants), 40 TypeScript SDK tests (including real Groth16 proof generation against 8-edge circuits in ∼640ms), and 5 billing E2E tests proving 100 sequential inference payments against ShieldedCredits.

The RLN Mode circuit composition (`circuits/rln-payment/`) and settlement contract (`RLNSettlement.sol`) are included as working implementations. The composed circuit adds ∼80 constraints to the audited sub-circuits.

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